

SRM VEC

SRM Nagar, Kattankulathur – 603 203

DEPARTMENT OF AIDS

QUESTION BANK

SUBJECT: AD3445 – AI1

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UNIT I – INTRODUCTION AND PROBLEM SOLVING

Foundations of AI, Intelligent Agents, Environments, Problem-solving agents, Search spaces, Uninformed search strategies: Breadth-first search, Depth-first search, Uniform cost search, Iterative deepening, Bi-directional search

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Artificial Intelligence in terms of acting rationally.	BTL1	Remembering
2	What is a rational agent?	BTL1	Remembering
3	List the four components of a PEAS description for an intelligent agent.	BTL1	Remembering
4	Distinguish between a simple reflex agent and a model-based reflex agent.	BTL2	Understanding
5	What is the primary difference between a goal-based agent and a utility-based agent?	BTL2	Understanding
6	Define 'Task Environment' in the context of AI.	BTL1	Remembering
7	What is the difference between a discrete and a continuous environment?	BTL2	Understanding
8	Define the term 'State Space' in problem-solving.	BTL1	Remembering
9	What are the four criteria used to evaluate the performance of a search algorithm?	BTL1	Remembering
10	State the time and space complexity of Breadth-First Search (BFS).	BTL1	Remembering
11	Why is Depth-First Search (DFS) considered to be not optimal?	BTL2	Understanding
12	Under what condition is Uniform Cost Search (UCS) guaranteed to be optimal?	BTL2	Understanding
13	What is the main advantage of Iterative Deepening Search over Depth-First Search?	BTL2	Understanding
14	Define 'Completeness' in the context of search strategies.	BTL1	Remembering
15	What is a 'Transition Model' in a well-defined problem?	BTL1	Remembering
16	Differentiate between an episodic and a sequential environment.	BTL2	Understanding
17	How does Bi-directional search reduce the time complexity of a search process?	BTL2	Understanding
18	What is a 'Static Environment'?	BTL1	Remembering
19	Define the branching factor 'b' and depth 'd' in a search tree.	BTL1	Remembering
20	What is the role of a 'Goal Test' in a problem-solving agent?	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Examine the foundations of Artificial Intelligence by discussing the influence of Philosophy, Mathematics, and Economics on its development.	(16)	BTL4	Analyzing
2	Explain the four basic types of agent programs with neat architectural diagrams and illustrative examples for each.	(16)	BTL4	Analyzing
3	Specify the PEAS description and characterize the environment for an Automated Taxi Driver and a Satellite Image Analysis System.	(16)	BTL3	Applying
4	Compare and contrast the different properties of task environments, such as Observable vs. Partially Observable and Deterministic vs. Stochastic, using real-world scenarios.	(16)	BTL4	Analyzing
5	Describe the general structure of a problem-solving agent and demonstrate its application in solving the 8-puzzle problem.	(16)	BTL3	Applying
6	Analyze the Breadth-First Search (BFS) algorithm. Prove its completeness and optimality when all step costs are equal.	(16)	BTL4	Analyzing
7	Explain the Depth-First Search (DFS) algorithm. Discuss its behavior in infinite state spaces and suggest modifications to make it more robust.	(16)	BTL4	Analyzing

8	Evaluate the Uniform Cost Search (UCS) algorithm. Provide a step-by-step trace of UCS on a weighted graph and justify its optimality.	(16)	BTL5	Evaluating
9	Discuss the Iterative Deepening Depth-First Search (IDS) algorithm. Explain how it combines the benefits of BFS and DFS while managing memory efficiently.	(16)	BTL4	Analyzing
10	Elaborate on the Bi-directional Search strategy. Discuss the theoretical reduction in complexity and the practical challenges of implementing it.	(16)	BTL4	Analyzing
11	Design a state-space representation for the 'Water Jug Problem' (4L and 3L jugs) and outline the sequence of operators to reach a goal of 2L in the 4L jug.	(16)	BTL6	Creating
12	Critically evaluate the trade-offs between space and time complexity for BFS, DFS, and Iterative Deepening Search in a tree with a large branching factor.	(16)	BTL5	Evaluating
13	Construct a search tree for the Vacuum-World problem and explain how a model-based reflex agent would navigate it compared to a simple reflex agent.	(16)	BTL6	Creating
14	Explain the concept of 'Search Space' and differentiate between a search tree and a state-space graph. Why is this distinction critical for algorithm design?	(16)	BTL4	Analyzing
15	Develop a comprehensive comparison table for all uninformed search strategies (BFS, DFS, UCS, IDS, Bi-directional) based on time, space, completeness, and optimality.	(16)	BTL6	Creating
16	Apply the Uniform Cost Search algorithm to find the shortest path in a grid-based navigation problem where different terrains have varying movement costs.	(16)	BTL3	Applying
17	Critically analyze the 'Turing Test' as a definition of intelligence. What are its limitations in the context of modern AI research?	(16)	BTL5	Evaluating
18	Explain the five components of a well-defined problem (Initial state, Actions, Transition model, Goal test, Path cost) using the 'Missionaries and Cannibals' problem.	(16)	BTL3	Applying
19	Discuss the concept of 'Rationality' in AI. How does the performance measure influence the behavior of a rational agent in a dynamic environment?	(16)	BTL4	Analyzing
20	Design an intelligent agent for a 'Smart Warehouse Robot.' Define its PEAS, environment characteristics, and the most suitable agent architecture for this task.	(16)	BTL6	Creating

UNIT II – HEURISTIC SEARCH AND GAME PLAYING

Informed search: Greedy best-first search, A search, AO* search, Memory-bounded heuristic search, Local search: Hill climbing, Simulated annealing, Genetic algorithms, Adversarial search: Minimax algorithm, Alpha-beta pruning*

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a heuristic function and provide an example for the 8-puzzle problem.	BTL1	Remembering
2	Distinguish between informed search and uninformed search strategies.	BTL2	Understanding
3	State the evaluation function used in Greedy Best-First Search.	BTL1	Remembering
4	What are the two conditions required for A* search to be optimal?	BTL1	Remembering
5	Explain the concept of heuristic consistency (monotonicity).	BTL2	Understanding
6	Define an AND-OR graph and its relevance to AO* search.	BTL1	Remembering
7	Identify the primary limitation of the A* search algorithm regarding memory usage.	BTL2	Understanding
8	What is the 'Local Maxima' problem in Hill Climbing search?	BTL2	Understanding
9	Define the term 'Annealing Schedule' in the context of Simulated Annealing.	BTL1	Remembering
10	List the basic genetic operators used in Genetic Algorithms.	BTL1	Remembering
11	Explain the role of the 'Fitness Function' in Genetic Algorithms.	BTL2	Understanding
12	Define a Zero-sum game in the context of adversarial search.	BTL1	Remembering
13	What is the purpose of the Alpha and Beta values in Alpha-Beta pruning?	BTL2	Understanding
14	State the minimax decision rule for a two-player game.	BTL1	Remembering
15	How does the branching factor affect the efficiency of Alpha-Beta pruning?	BTL2	Understanding
16	Define 'Stochastic Hill Climbing' and how it differs from simple Hill Climbing.	BTL1	Remembering
17	What is the significance of the 'Temperature' parameter in Simulated Annealing?	BTL2	Understanding
18	Compare 'Crossover' and 'Mutation' in Genetic Algorithms.	BTL2	Understanding
19	Name two memory-bounded heuristic search algorithms.	BTL1	Remembering
20	Explain why Alpha-Beta pruning does not affect the final decision of the Minimax algorithm.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the A* search algorithm in detail. Illustrate its working with a numerical example and prove that A* is optimal if the heuristic is admissible.	(16)	BTL3	Applying
2	Compare and contrast Greedy Best-First Search and A* Search. Analyze their performance in terms of completeness, optimality, and complexity with suitable diagrams.	(16)	BTL4	Analyzing
3	Describe the AO* algorithm for searching AND-OR graphs. Discuss how it handles the cost of solution trees and provide a step-by-step trace on a sample graph.	(16)	BTL4	Analyzing
4	Analyze the Recursive Best-First Search (RBFS) and Simplified Memory-bounded A* (SMA*) algorithms. Explain how they manage memory constraints while maintaining search efficiency.	(16)	BTL4	Analyzing
5	Evaluate the different variants of Hill Climbing search. Discuss the strategies used to overcome problems like ridges, plateaus, and local maxima.	(16)	BTL5	Evaluating
6	Explain the Simulated Annealing algorithm. Discuss how the probability of accepting a worse move changes over time and the impact of different cooling schedules.	(16)	BTL4	Analyzing

7	Design a Genetic Algorithm to solve the 8-Queens problem. Clearly define the chromosome representation, fitness function, selection method, crossover, and mutation operators.	(16)	BTL6	Creating
8	Explain the Minimax algorithm for adversarial search. Illustrate its execution on a game tree of depth 3 and calculate the minimax values at each node.	(16)	BTL3	Applying
9	Derive the Alpha-Beta pruning procedure. Demonstrate with a complex game tree how pruning reduces the number of states explored and discuss the best-case time complexity.	(16)	BTL5	Evaluating
10	Discuss the concept of 'Heuristic Dominance'. If $h_2(n) \geq h_1(n)$ for all n , analyze why A* using h_2 is more efficient than A* using h_1 .	(16)	BTL4	Analyzing
11	Compare Hill Climbing, Simulated Annealing, and Genetic Algorithms as local search techniques. Justify which algorithm is best suited for large-scale combinatorial optimization.	(16)	BTL5	Evaluating
12	Formulate a heuristic search approach for the Traveling Salesperson Problem (TSP). Propose an admissible heuristic and explain how A* would solve it.	(16)	BTL6	Creating
13	Analyze the impact of the 'Horizon Effect' in game playing. Explain how techniques like Quiescence search and Singular extensions help in mitigating this problem.	(16)	BTL4	Analyzing
14	Explain the IDA* (Iterative Deepening A*) algorithm. Compare its space and time complexity with standard A* and Iterative Deepening Depth-First Search.	(16)	BTL4	Analyzing
15	Design an AND-OR graph representation for a complex robot planning task and apply the AO* algorithm to find the optimal plan.	(16)	BTL6	Creating
16	Evaluate the effectiveness of different selection strategies in Genetic Algorithms, such as Roulette Wheel selection and Tournament selection, on population convergence.	(16)	BTL5	Evaluating
17	Apply the Minimax algorithm with Alpha-Beta pruning to a Tic-Tac-Toe game state and show the pruned branches clearly.	(16)	BTL3	Applying
18	Discuss the properties of heuristic functions. Explain how to construct an admissible heuristic from subproblems using pattern databases.	(16)	BTL4	Analyzing
19	Critically analyze the trade-offs between time complexity and space complexity in memory-bounded search algorithms like MA* and SMA*.	(16)	BTL5	Evaluating
20	Develop a search-based solution for the 'Missionaries and Cannibals' problem using the A* algorithm, including the definition of state space and a valid heuristic.	(16)	BTL6	Creating

UNIT III – KNOWLEDGE REPRESENTATION AND LOGIC

Knowledge-based agents, Propositional logic, First-order logic: Syntax and semantics, Knowledge engineering, Inference in FOL: Unification, Forward and backward chaining, Resolution, Ontological engineering

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a Knowledge-based agent and its primary components.	BTL1	Remembering
2	Distinguish between 'Entailment' and 'Inference' in logical systems.	BTL2	Understanding
3	What is meant by a 'Sound' inference algorithm?	BTL1	Remembering
4	Define the 'Completeness' property of a logic system.	BTL1	Remembering
5	Explain the significance of 'Validity' and 'Satisfiability' in Propositional Logic.	BTL2	Understanding
6	Convert the English sentence 'Every student loves AI' into a First-order Logic expression.	BTL2	Understanding
7	What are the two types of quantifiers used in First-order Logic?	BTL1	Remembering
8	Define 'Unification' in the context of First-order Logic inference.	BTL1	Remembering
9	What is a 'Most General Unifier' (MGU)?	BTL1	Remembering
10	Briefly explain the 'Closed World Assumption'.	BTL2	Understanding
11	State the basic rule of 'Resolution' for Propositional Logic.	BTL1	Remembering
12	What is a 'Horn Clause' and why is it important for inference?	BTL2	Understanding
13	Define 'Knowledge Engineering' and its role in AI.	BTL1	Remembering
14	List the steps involved in the Knowledge Engineering process.	BTL1	Remembering
15	Explain the purpose of 'Skolemization' in First-order Logic.	BTL2	Understanding
16	What is an 'Ontology' in the context of AI?	BTL1	Remembering
17	Differentiate between 'Forward Chaining' and 'Backward Chaining' at a high level.	BTL2	Understanding
18	What is 'Ontological Engineering'?	BTL1	Remembering
19	Define 'Atomic Sentences' in First-order Logic.	BTL1	Remembering
20	Explain the role of a 'Substitution' in logical unification.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Design a Knowledge-based agent for the Wumpus World environment. Explain its architecture, the representation of its knowledge base, and how it performs reasoning to navigate safely.	(16)	BTL6	Creating
2	Compare and contrast Propositional Logic and First-order Logic with respect to their syntax, semantics, and expressive power. Provide examples to support your analysis.	(16)	BTL4	Analyzing
3	Given the following facts: 'John likes all kinds of food. Apples are food. Chicken is food. Anything anyone eats and isn't killed by is food. Bill eats peanuts and is still alive. Sue eats everything Bill eats.' Use Resolution Refutation to prove that 'John likes peanuts'.	(16)	BTL3	Applying
4	Explain the detailed procedure for converting a First-order Logic formula into Conjunctive Normal Form (CNF). Illustrate each step using a complex logical expression.	(16)	BTL3	Applying
5	Describe the Forward Chaining algorithm for First-order Logic. Illustrate its execution with a suitable example involving a set of rules and facts, and discuss its efficiency.	(16)	BTL3	Applying
6	Evaluate the advantages and disadvantages of Backward Chaining compared to Forward Chaining in goal-directed search scenarios. In which situations is one preferred over the other?	(16)	BTL5	Evaluating

7	Develop a Knowledge Engineering framework for building an expert system in the domain of 'Automobile Fault Diagnosis'. Outline the steps from knowledge acquisition to verification.	(16)	BTL6	Creating
8	Analyze the Unification algorithm in First-order Logic. Provide a detailed pseudo-code and demonstrate its application on several pairs of expressions, including cases where unification fails.	(16)	BTL4	Analyzing
9	Critically examine the 'Frame Problem' in knowledge representation. Explain why it is a challenge for logic-based systems and discuss potential solutions like Successor-State Axioms.	(16)	BTL4	Analyzing
10	Construct an ontology for a 'Smart University Management System'. Identify the key classes, subclasses, properties, and axioms required to represent students, courses, and faculty relationships.	(16)	BTL6	Creating
11	Prove the validity of the hypothetical syllogism $((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$ using both Truth Table method and Natural Deduction rules.	(16)	BTL3	Applying
12	Discuss the concept of 'Upper Ontologies' (like Cyc or SUMO). Evaluate their importance in achieving semantic interoperability between diverse AI systems and knowledge bases.	(16)	BTL5	Evaluating
13	Analyze the semantics of First-order Logic by explaining the roles of Interpretations, Models, and Variable Assignments in determining the truth of a formula.	(16)	BTL4	Analyzing
14	Compare 'Model Checking' and 'Theorem Proving' as approaches for logical inference. Discuss their computational complexity and suitability for large-scale knowledge bases.	(16)	BTL4	Analyzing
15	Demonstrate the use of Resolution to show that a given set of clauses is unsatisfiable. Explain why Resolution is considered a refutation-complete inference procedure.	(16)	BTL3	Applying
16	Critically evaluate the use of Logic Programming (e.g., Prolog) for solving complex reasoning problems. How does it handle the trade-off between expressiveness and efficiency?	(16)	BTL5	Evaluating
17	Design a set of First-order Logic sentences to represent a 'Family Tree' (including relations like parent, sibling, grandparent, and cousin). Use these to derive a new relationship using inference.	(16)	BTL6	Creating
18	Analyze the role of 'Categories' and 'Measures' in Ontological Engineering. How are abstract concepts like 'Time' and 'Space' typically represented in a formal ontology?	(16)	BTL4	Analyzing
19	Explain the 'Resolution Refutation' procedure for FOL. Why must we use Skolemization and Unification as prerequisites for this process?	(16)	BTL5	Evaluating
20	Investigate the limitations of First-order Logic in representing 'Common Sense Knowledge'. Discuss how Ontological Engineering attempts to bridge the gap between formal logic and human-like reasoning.	(16)	BTL4	Analyzing

UNIT IV – REASONING AND UNCERTAINTY

Quantifying uncertainty, Probabilistic reasoning, Bayesian networks: Semantics and inference, Exact and approximate inference, Temporal models: Hidden Markov Models, Kalman filters, Decision networks

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define uncertainty in the context of Artificial Intelligence and list its primary sources.	BTL1	Remembering
2	State the three fundamental axioms of probability.	BTL1	Remembering
3	Explain the significance of Bayes' Rule in probabilistic reasoning.	BTL2	Understanding
4	What is a Bayesian Network and what does it represent?	BTL1	Remembering
5	Define a Conditional Probability Table (CPT) and its role in Bayesian Networks.	BTL1	Remembering
6	Explain the concept of d-separation in a directed acyclic graph.	BTL2	Understanding
7	What constitutes the Markov Blanket of a node in a Bayesian Network?	BTL1	Remembering
8	Distinguish between exact and approximate inference methods.	BTL2	Understanding
9	What is the core idea behind the Variable Elimination algorithm?	BTL2	Understanding
10	List two common sampling techniques used for approximate inference.	BTL1	Remembering
11	Define a Hidden Markov Model (HMM) and its key components.	BTL1	Remembering
12	Explain the 'Filtering' task in the context of temporal models.	BTL2	Understanding
13	What is the 'Smoothing' task in Hidden Markov Models?	BTL2	Understanding
14	State the purpose of the Viterbi algorithm in temporal reasoning.	BTL2	Understanding
15	Define a Kalman Filter and specify when it is preferred over an HMM.	BTL1	Remembering
16	What is a Decision Network (Influence Diagram)?	BTL1	Remembering
17	Define the Principle of Maximum Expected Utility (MEU).	BTL1	Remembering
18	Differentiate between state variables and evidence variables in temporal models.	BTL2	Understanding
19	Explain the 'Stationary Process' assumption in Markov chains.	BTL2	Understanding
20	What is Likelihood Weighting and how does it improve upon Rejection Sampling?	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the semantics of Bayesian Networks and derive the general formula for the joint probability distribution of a set of variables.	(16)	BTL3	Applying
2	Describe the Variable Elimination algorithm in detail and demonstrate its execution on a sample network to compute a marginal distribution.	(16)	BTL3	Applying
3	Analyze the differences between Rejection Sampling and Likelihood Weighting, evaluating their performance in networks with rare evidence.	(16)	BTL4	Analyzing
4	Explain the Gibbs Sampling algorithm for approximate inference and discuss its convergence properties in high-dimensional spaces.	(16)	BTL5	Evaluating
5	Design a Bayesian Network for a diagnostic system (e.g., automotive engine failure) including at least five nodes, and justify the conditional independence assumptions made.	(16)	BTL6	Creating
6	Discuss the four basic inference tasks in temporal models: Filtering, Prediction, Smoothing, and Finding the most likely sequence, with relevant mathematical formulations.	(16)	BTL4	Analyzing
7	Derive the Forward-Backward algorithm for Hidden Markov Models and explain how it is utilized for the smoothing task.	(16)	BTL4	Analyzing

8	Explain the mathematical foundations of Kalman Filters, focusing on the prediction and update cycles for continuous state variables.	(16)	BTL3	Applying
9	Compare and contrast Hidden Markov Models and Kalman Filters in terms of state representation, transition models, and typical application domains.	(16)	BTL4	Analyzing
10	Design a Decision Network for a medical treatment scenario, incorporating chance nodes for disease, action nodes for treatment, and utility nodes for patient outcomes.	(16)	BTL6	Creating
11	Evaluate the impact of the first-order Markov Assumption on the computational complexity and representational accuracy of temporal reasoning systems.	(16)	BTL5	Evaluating
12	Explain the concept of 'Value of Information' and demonstrate how it can be calculated using a decision network to prioritize data collection.	(16)	BTL4	Analyzing
13	Discuss the semantics of d-separation by illustrating the three basic structural patterns: causal chains, common causes, and common effects (v-structures).	(16)	BTL3	Applying
14	Analyze why exact inference in Bayesian Networks is NP-hard and justify the necessity of using Monte Carlo Markov Chain (MCMC) methods for large-scale problems.	(16)	BTL4	Analyzing
15	Explain the process of performing inference in temporal models using Particle Filtering, highlighting its advantages for non-linear systems.	(16)	BTL3	Applying
16	Develop a complete Hidden Markov Model for a Part-of-Speech (POS) tagging application, defining the state space, transition matrix, and emission probabilities.	(16)	BTL6	Creating
17	Critically evaluate the use of Bayesian Networks versus traditional Rule-based Expert Systems for handling uncertainty in real-world engineering applications.	(16)	BTL5	Evaluating
18	Explain how Bayesian Networks can be used for 'Parameter Learning' when the network structure is fixed but the CPT values are unknown, using the Maximum Likelihood Estimation approach.	(16)	BTL3	Applying
19	Discuss the 'Most Likely Explanation' problem in HMMs and provide a detailed algorithmic walkthrough of the Viterbi algorithm with a numerical example.	(16)	BTL4	Analyzing
20	Design a probabilistic reasoning framework for an autonomous drone navigation system using a combination of Kalman Filters for localization and Decision Networks for path planning.	(16)	BTL6	Creating

UNIT V – PLANNING AND EXPERT SYSTEMS

Classical planning, Planning as state-space search, Planning graphs, Hierarchical planning, Multi-agent planning, Introduction to Expert Systems, Rule-based systems, AI applications in Robotics and NLP

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define classical planning in the context of Artificial Intelligence.	BTL1	Remembering
2	What are the three main components of a PDDL action schema?	BTL1	Remembering
3	Distinguish between progression and regression planning in state-space search.	BTL2	Understanding
4	What is a Planning Graph and what purpose does it serve?	BTL1	Remembering
5	List the three types of mutex constraints used in planning graphs.	BTL1	Remembering
6	Explain the concept of Hierarchical Task Network (HTN) planning.	BTL2	Understanding
7	What is the primary difference between single-agent and multi-agent planning?	BTL2	Understanding
8	Define an Expert System and name its core components.	BTL1	Remembering
9	What is the role of the Knowledge Base in a rule-based system?	BTL1	Remembering
10	Explain the term 'Conflict Resolution' in the context of production systems.	BTL2	Understanding
11	Mention two specific AI applications in the field of Robotics.	BTL1	Remembering
12	How does AI planning contribute to Natural Language Processing (NLP)?	BTL2	Understanding
13	What is the 'Closed World Assumption' in classical planning?	BTL1	Remembering
14	Define 'Heuristic Search' as applied to planning problems.	BTL1	Remembering
15	What are non-primitive tasks in Hierarchical Planning?	BTL2	Understanding
16	Explain the concept of 'Joint Intentions' in multi-agent systems.	BTL2	Understanding
17	What is the function of an Inference Engine?	BTL1	Remembering
18	Differentiate between Forward Chaining and Backward Chaining.	BTL2	Understanding
19	What does it mean when a planning graph 'levels off'?	BTL1	Remembering
20	Define the 'STRIPS' operator and its elements.	BTL1	Remembering

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the STRIPS representation for planning in detail and demonstrate its application using the Blocks World problem with a specific start and goal state.	(16)	BTL3	Applying
2	Analyze the efficiency and search space complexity of Forward State-Space Search versus Backward State-Space Search with suitable examples.	(16)	BTL4	Analyzing
3	Construct a Planning Graph for the 'Spare Tire' problem and identify all mutex relations (Inconsistent Effects, Interference, and Competing Needs) at each level.	(16)	BTL6	Creating
4	Evaluate the Graphplan algorithm's approach to handling conjunctive goals and discuss how it extracts a valid plan from the graph.	(16)	BTL5	Evaluating
5	Design a Hierarchical Task Network (HTN) for a 'Travel Planning' scenario, detailing the decomposition of non-primitive tasks into primitive actions.	(16)	BTL6	Creating
6	Discuss the challenges involved in Multi-agent Planning, focusing on coordination protocols and communication strategies in a decentralized environment.	(16)	BTL4	Analyzing
7	Explain the architecture of a typical Expert System and justify the importance of the Explanation Facility and User Interface in its adoption.	(16)	BTL5	Evaluating

8	Develop a rule-based system for a 'Medical Diagnosis' application, including at least eight rules, and explain the inference mechanism used to reach a conclusion.	(16)	BTL6	Creating
9	Compare and contrast Forward Chaining and Backward Chaining inference techniques, specifying the conditions under which one is preferred over the other.	(16)	BTL4	Analyzing
10	Analyze the role of AI in Robotics, specifically focusing on the integration of path planning, sensor fusion, and obstacle avoidance algorithms.	(16)	BTL4	Analyzing
11	Evaluate the application of AI planning techniques in Natural Language Processing for building task-oriented dialogue systems.	(16)	BTL5	Evaluating
12	Illustrate the process of 'Heuristic Generation' from relaxed problems in classical planning and explain how the h-max and h-add heuristics are calculated.	(16)	BTL3	Applying
13	Explain the 'Partial-Order Planning' approach and compare its flexibility and performance with Total-Order Planning techniques.	(16)	BTL4	Analyzing
14	Design a multi-agent coordination mechanism for a warehouse automation system where multiple robots must pick and pack items without collision.	(16)	BTL6	Creating
15	Critically examine the limitations of traditional Rule-Based Systems and suggest how modern Expert Systems incorporate uncertainty using fuzzy logic or probability.	(16)	BTL5	Evaluating
16	Apply the concept of 'State-Space Search' to solve the 8-puzzle problem using the A* algorithm and justify the choice of an admissible heuristic.	(16)	BTL3	Applying
17	Discuss the integration of Natural Language Processing and Expert Systems in building intelligent Virtual Assistants for customer support.	(16)	BTL4	Analyzing
18	Formulate a PDDL description for a 'Logistics Domain' involving trucks, cities, and packages, including predicates, actions, and a sample problem instance.	(16)	BTL6	Creating
19	Assess the impact of 'Sensing and Execution Monitoring' in real-world robotic planning where the environment is partially observable and dynamic.	(16)	BTL5	Evaluating
20	Analyze the performance of the Rete Algorithm in large-scale rule-based systems and explain how it optimizes the pattern-matching process.	(16)	BTL4	Analyzing

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